

Prateek Aryan

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SUMMARY

Senior Software Engineer with 5+ years designing and scaling distributed systems, backend platforms, and production LLM/GenAI pipelines. Currently building fault-tolerant, cost-optimized RAG and agent-orchestration systems for regulated clinical-document generation. Deep expertise in resiliency-focused system design, databases, and DevOps automation supporting 200+ enterprise applications, partnering with Product, Data Science, and domain teams.

TECHNICAL SKILLS

Languages: Python, JavaScript

AI / LLM Engineering: RAG pipeline design, prompt caching, model routing & tiering, durable agent orchestration, LLM observability & evaluation, AWS Bedrock (Claude), Anthropic API, OpenSearch vector retrieval

Frameworks: Django, Flask, FastAPI, Express.js, React, Angular, REST APIs, gRPC

Databases & Messaging: PostgreSQL, MongoDB, MySQL, Oracle, Redis, Cassandra, Kafka

DevOps & Cloud: AWS (Bedrock, EKS, ECR, RDS, S3, Lambda, Step Functions), Docker, Kubernetes, Terraform, Ansible, Jenkins, Helm, CI/CD

System Design: Distributed Systems, Microservices, Resiliency & Reliability, Scalability

WORK EXPERIENCE

UsefulBI

Bengaluru

Senior Software Engineer II / AI Engineer

Nov 2025 – Present

Gilead Sciences (client) — production clinical content-generation platform, CSR/DSUR/PLPS:

- Designed a unified orchestration layer for LLM document generation using **durable execution with a Postgres-backed checkpointer** — isolating low-latency interactive requests from long-running generations and guaranteeing **zero state loss** across worker restarts and failures.
- Engineered an **idempotent, multi-stage RAG pipeline** (extract → chunk → embed → retrieve → generate) with **content-hash-keyed caching** at each stage, skipping unchanged work to cut repeat generations **from minutes to seconds**.
- Reduced inference latency **~85%** and API token cost **~90%** on high-volume RAG instructions by introducing **prompt caching** for stable system prefixes.
- Built a **model-tiering router** that classifies intent with fast, low-cost models under **strict Pydantic schemas** before escalating complex generation to frontier models — protecting both cost and output quality.
- Architected a **two-layer LLM observability pipeline** pairing offline dataset scoring with live trace transformation, enabling **drift detection and automated quality alerting** in production.
- UsefulBI clinical-document POC (sole developer — the demo that won the Gilead engagement):**
- Built and **solely owned** UsefulBI's AI clinical-document platform — FastAPI microservices and a 5-stage RAG pipeline (Textextract → chunk → Titan-v2 embeddings on OpenSearch → Bedrock/Claude generation → DOCX with citations), with section-level versioning, a 4-level review/approval cascade, and audit/RBAC **designed for 21 CFR Part 11**.
- Cut clinical-document retrieval from **7–8s to under 50ms (99.3%)** by replacing synchronous S3 polling with an indexed database registry (**O(n) → O(1)**), and stood up the team's first **EKS/ECR/Helm CI/CD** pipeline — taking deploys from hours of manual work to under 15 minutes.

Cisco Systems

Bengaluru

Software Engineer

Feb 2021 – Nov 2025

- Built **large-scale automation for database operations** — MySQL upgrades, Postgres provisioning, Oracle switchover and patching, production-to-nonprod refreshes — supporting **200+ enterprise applications** and reducing manual DBA workload by **80%**.
- Built a resilient switchover automation system using **gRPC** for low-latency communication across primary and standby **distributed clusters**, ensuring system resiliency and reliability at scale.
- Designed and implemented database provisioning workflows using **Terraform, Flask, and Ansible** — automating VM creation on OpenStack, instance configuration, and production data refresh for non-production environments.
- Accelerated SOX compliance reporting from **3+ hours to under 10 minutes** by building an automated reporting pipeline, analysing production features using analytics to measure operational impact.
- Managed self-hosted MongoDB clusters at scale — backup/recovery, replication, and performance tuning for production workloads powering **100+ enterprise applications**. Initially joined as intern; automated MySQL cluster upgrades, cutting time from **1 hour to 10 minutes**.

EDUCATION

BMS College of Engineering

B.E. in Computer Science — GPA: 9.2

2017 – 2021

CERTIFICATIONS

MongoDB Certified DBA | AWS Certified Cloud Practitioner | HashiCorp Terraform Associate | PCAP — Python Programming